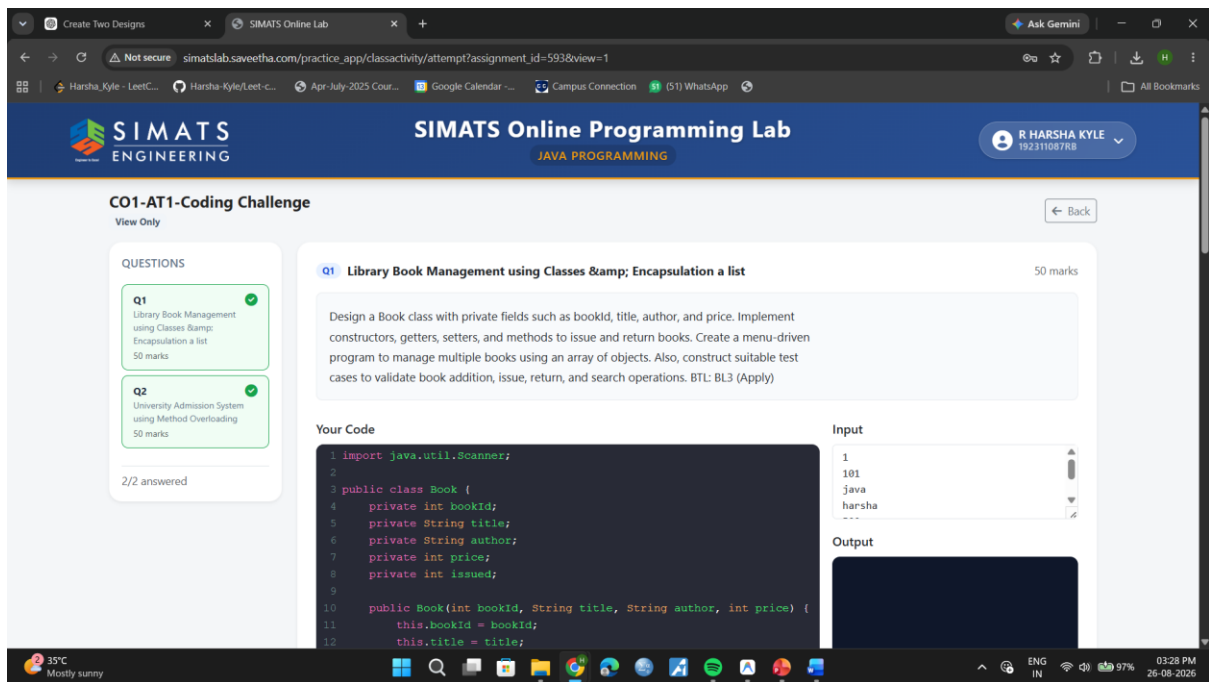


1.



```
import java.util.Scanner;
```

```
public class Book {
    private int bookId;
    private String title;
    private String author;
    private int price;
    private int issued;
```

```
    public Book(int bookId, String title, String author, int price) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.price = price;
        this.issued = 0;
    }
```

```
    public int getBookId() {
        return bookId;
    }
```

```
    public void issueBook() {
        if (issued == 0) {
            issued = 1;
```

```
System.out.println("Issued: " + title);
} else {
System.out.println("Already issued: " + title);
}
}
```

```
public void returnBook() {
if (issued == 1) {
issued = 0;
System.out.println("Returned: " + title);
} else {
System.out.println("Not issued: " + title);
}
}
```

```
public void display() {
String s = (issued == 1) ? "Issued" : "Available";
System.out.println(bookId + " " + title + " " + author + " " + price + " " + s);
}
```

```
static Book[] books = new Book[100];
static int count = 0;
```

```
static void addBook(Scanner sc) {
int id = Integer.parseInt(sc.nextLine());
String t = sc.nextLine();
String a = sc.nextLine();
int p = Integer.parseInt(sc.nextLine());
books[count++] = new Book(id, t, a, p);
System.out.println("Book added.");
}
```

```
static Book search(int id) {
for (int i = 0; i < count; i++) {
if (books[i].getBookId() == id) return books[i];
}
return null;
}
```

```
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
```

```

int choice;

do {
    System.out.println("1.Add 2.Display 3.Issue 4.Return 5.Search 6.Exit");
    choice = Integer.parseInt(sc.nextLine());

    if (choice == 1) {
        addBook(sc);
    } else if (choice == 2) {
        for (int i = 0; i < count; i++) books[i].display();
    } else if (choice == 3) {
        Book b = search(Integer.parseInt(sc.nextLine()));
        if (b != null) b.issueBook();
        else System.out.println("Not found");
    } else if (choice == 4) {
        Book b = search(Integer.parseInt(sc.nextLine()));
        if (b != null) b.returnBook();
        else System.out.println("Not found");
    } else if (choice == 5) {
        Book b = search(Integer.parseInt(sc.nextLine()));
        if (b != null) b.display();
        else System.out.println("Not found");
    } else if (choice == 6) {
        System.out.println("Exiting.");
    } else {
        System.out.println("Invalid.");
    }
} while (choice != 6);
}
}2

```

2.

The screenshot shows a web browser window with the SIMATS Online Programming Lab interface. The page title is "SIMATS Online Programming Lab" with a sub-header "JAVA PROGRAMMING". The user is logged in as "R HARSHA KYLE" with ID "192311087RB". The challenge is titled "CO1-AT1-Coding Challenge" and is in "View Only" mode. It lists two questions: "Q1 Library Book Management using Classes & Encapsulation a list 50 marks" and "Q2 University Admission System using Method Overloading 50 marks". The selected question, Q2, asks to create an Admission class that overloads the method calculateFee() for Undergraduate, Postgraduate, and Scholarship students, and to display the fee structure based on user input. The "Your Code" section shows a Java code snippet for the Main class. The "Input" section shows "Undergraduate", "Postgraduate", "true", and "5". The "Output" section is empty.

```
1 import java.util.Scanner;
2
3 public class Main {
4
5     static int calculatefee(String prgm) {
6         int fee;
7         if (prgm.equalsIgnoreCase("Undergraduate")) {
8             fee = 350000;
9         } else if (prgm.equalsIgnoreCase("Postgraduate")) {
10            fee = 500000;
11        } else {
12            fee = 0;
13        }
14        return fee;
15    }
16}
```

```
import java.util.Scanner;
```

```
public class Main {
```

```
    static int calculatefee(String prgm) {
        int fee;
        if (prgm.equalsIgnoreCase("Undergraduate")) {
            fee = 350000;
        } else if (prgm.equalsIgnoreCase("Postgraduate")) {
            fee = 500000;
        } else {
            fee = 0;
        }
        return fee;
    }
}
```

```
    static int calculatefee(String prgm, boolean sch) {
        int fee;
```

```
        if (prgm.equalsIgnoreCase("Undergraduate")) {
            if (sch) {
                fee = 350000;
            } else {
                fee = 500000;
            }
        }
    }
}
```

```

    }
    } else if (prgm.equalsIgnoreCase("Postgraduate")) {
    if (sch) {
    fee = 350000;
    } else {
    fee = 500000;
    }
    } else {
    fee = 0;
    }

    return fee;
}

static int calculatefee(int sem) {
int oneSemFee = 120000;
return sem * oneSemFee;
}

public static void main(String[] args) {
Scanner sc = new Scanner(System.in);

System.out.println("Enter Program Type (Undergraduate/Postgraduate):");
String p1 = sc.nextLine();
System.out.println("Fee is: " + calculatefee(p1));

System.out.println("Enter Program Type (Undergraduate/Postgraduate):");
String p2 = sc.nextLine();
System.out.println("Scholarship (true/false):");
boolean s = sc.nextBoolean();
System.out.println("Fee is: " + calculatefee(p2, s));

System.out.println("Enter Number of Semesters:");
int sem = sc.nextInt();
System.out.println("Fee is: " + calculatefee(sem));
}
}

```